

Colorimetric Analysis Methodology

Bacterial–Dye–Antibiotic Arrays

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Problem Setup

- ▶ **Objective:** Extract biological insights from a temporal sequence of colorimetric images (0^{th} min to 25^{th} min).
- ▶ **System:** Bacterial–dye–antibiotic arrays (3x9 grid matrix).
- ▶ **Goal:** Determine the Minimum Inhibitory Concentration (MIC) and understand reaction kinetics.
- ▶ **Indicator:** Biological indicator dye transitions from blue (high a^*) to pink (higher a^*) as surviving bacteria reduce it in a redox reaction.

Literature Survey

▶ Freeze-Dried Cellulose for AST (2026)

- ▶ *Source:* Sarita C, Khandelwal M, Duraiswamy S. Freeze-Dried Bacterial Cellulose-Based Point-of-Care Device for Antibiotic Susceptibility Testing (AST) (2026).
- ▶ *Relevance:* Context, background, need.

▶ Smartphone-Based Concentration Classification

- ▶ *Source:* ACS Omega (2021).
- ▶ *Relevance:* Establishes that smartphone images can be used for “coarse” classification of chemical concentrations. This validates our approach of using computer vision on simple camera images rather than expensive spectrophotometers.

▶ ML-Assisted Chemiluminescence

- ▶ *Source:* Talanta (2023).
- ▶ *Relevance:* Highlights the power of Machine Learning (ML) in discriminating subtle signal changes (chemiluminescence) that human eyes might miss. We apply this same logic to subtle color shifts (Purple to Pink).

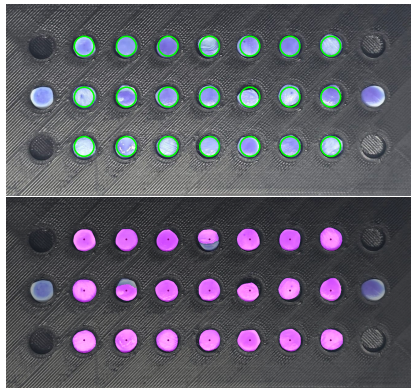
Grid Detection & Extraction

- ▶ **Histogram Projection:**

Isolate bright sample wells from the dark tray background using grayscale conversion, thresholding, and morphology opening.

- ▶ **Peak Finding:** Identify well centers via prominent peaks in projected 1D histograms over X and Y axes.

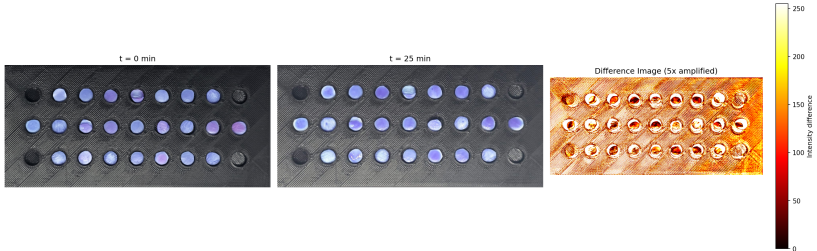
- ▶ **Median Extraction:** Draw a 32-pixel radius circular mask at each identified well location and extract median intensity across RGB, CIELAB ($L^*a^*b^*$), and HSV color spaces.



1. Visual Difference Analysis

- ▶ **Approach:** Compare first and last frame visually and via absolute difference map.
- ▶ **Reported result:** Mean absolute intensity difference was about 44.
- ▶ **Reported result:** About 4.2 million pixels changed by more than 20 intensity units.
- ▶ **Interpretation:** Visible change exists, and we need the pipelines to convert it into strong biological trend metrics.

Change Detection: Brighter areas = more change

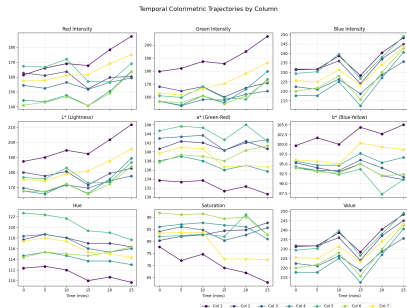


2. Circular ROI Median Color Extraction

- ▶ **Approach:** Detect grid peaks, place fixed-radius circular masks, extract median color per well.
- ▶ **Strength:** Robust to outliers compared with mean-based extraction.
- ▶ **Limitation:** Sensitive to misalignment when grid detection or geometry changes.
- ▶ **Result:** Stable replicate behavior (CV roughly 2% to 9%), but weak biological separation in a^* .

3. Temporal MIC Identification

- ▶ **Method:** Automated kinetic slope detection on the a^* (green-red) channel over time.
- ▶ **Threshold:** Monotonic linear regression slope ($m > 0.1$) indicates bacterial survival.
- ▶ **Result:** Column 5 identified as the MIC (last completely inhibited concentration).



4. Ratio-Normalized Metrics

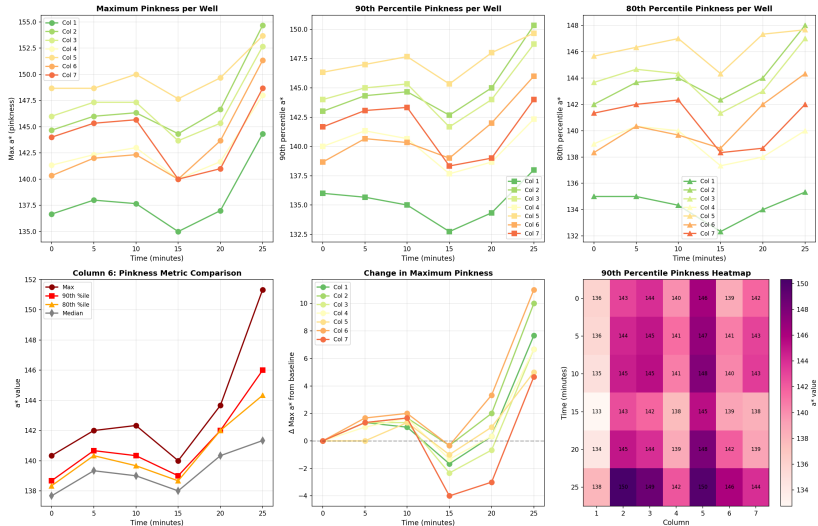
- ▶ **Approach:** Use R/G and A/L ratios to reduce illumination sensitivity.
- ▶ **R/G results:** Mostly small negative shifts; strongest positive shift in one column (+0.0530).
- ▶ **A/L results:** All columns negative (about -0.023 to -0.107).
- ▶ **Interpretation:** Normalization confirms broad brightening effects and weak concentration-ordered biological contrast.

5. Pinkness Percentile Analysis

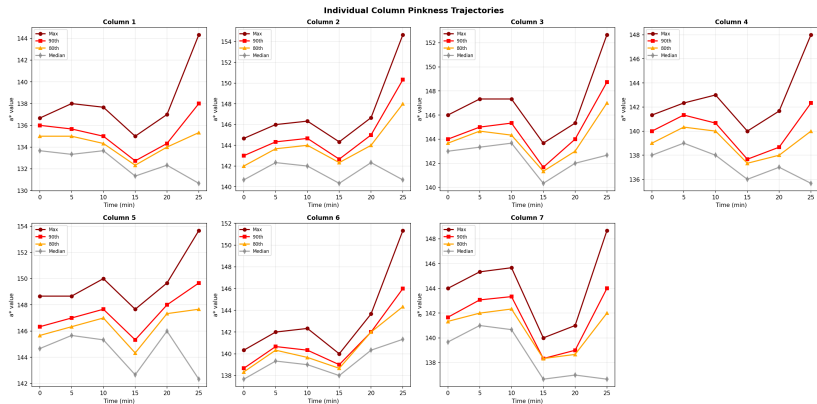
- ▶ **Approach:** Track upper-tail pink pixels ($a_{\max}^*$, a_{p90}^* , a_{p80}^*) rather than medians.
- ▶ **Observation:** There is correlation with time, but no correlation with the concentration.
- ▶ **Interpretation:** Changes are visible in each well across time, but not consistently across different wells.
- ▶ Also tested with R instead of a^* .

5. Pinkness Percentile Analysis Results

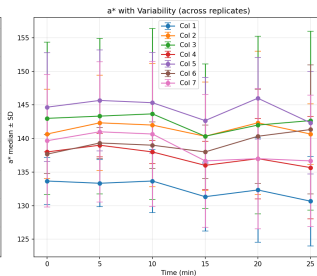
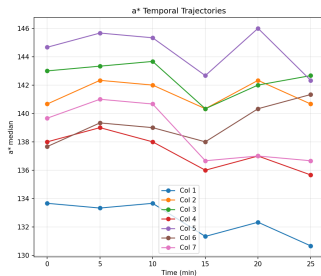
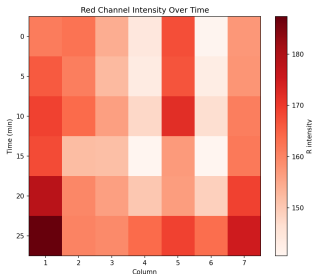
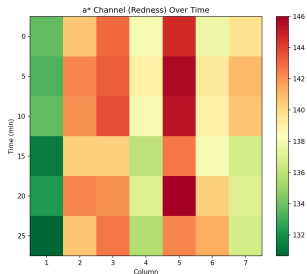
Temporal Pinkness Analysis: Max and Percentile Metrics



5. Pinkness Percentile Analysis Results

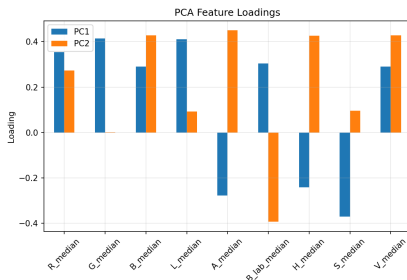
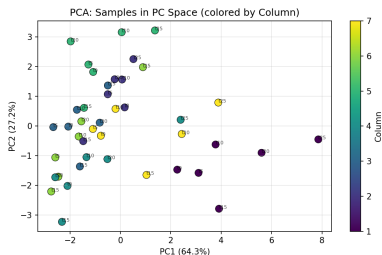


5. Pinkness Percentile Analysis Results



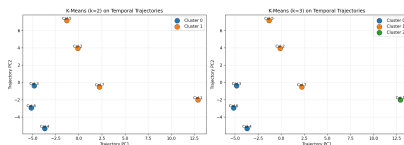
6. Principal Component Analysis (PCA)

- ▶ **Purpose:** Dimensionality reduction of 9 distinct color features per sample per timepoint.
- ▶ **Inference:** Green (G), Lightness (L^*), and Saturation (S) channels carry the highest variance over the temporal sequence, more than traditional red-blue axes.



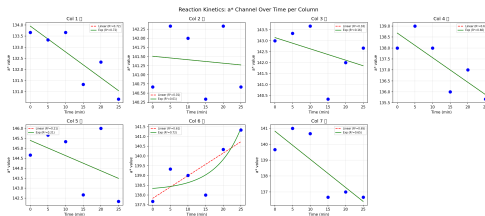
7. K-Means Behavioral Clustering

- ▶ **Method:** Clustering flattened temporal trajectories ($K = 3$).
- ▶ **Clusters identified:**
 1. High Inhibition (Col 1)
 2. Transition/Stable Zone (Cols 2, 5, 7)
 3. High Activity (Cols 3, 4, 6)
- ▶ **Inference:** Response is not a smooth gradient but forms discrete "plateaus" of efficacy.



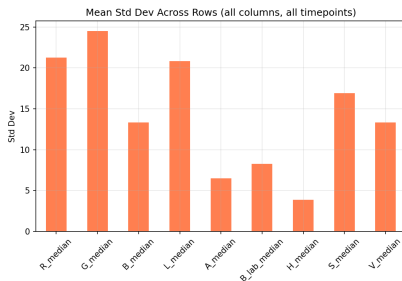
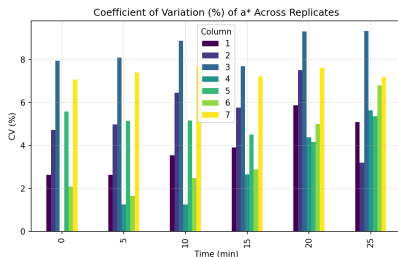
8. Early Prediction & Curvilinear Kinetics

- **Early Forecasting:** Logistic regression classifier achieved 85.7% accuracy predicting final MIC outcome at $t = 25$ using *only* the initial $t = 0$ image.
- **Curvilinear Kinetics:** Extracting R^2 exponential decay/growth coefficients using non-linear curve fits to overlying a^* channel datapoints. Provides insight into deceleration rate and specific bacteriostatic/bactericidal mechanism.



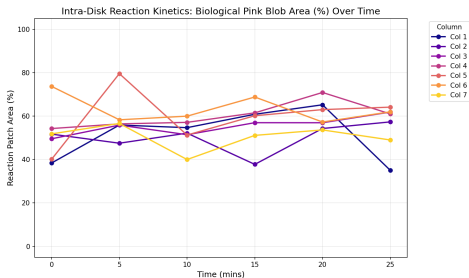
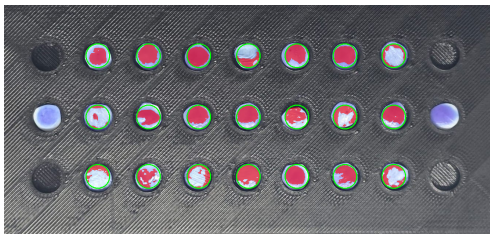
9. Replicate Consistency

- ▶ **Method:** Calculated Coefficient of Variation (CV%) across the 3 replicate rows.
- ▶ **Inference:** Assay layout is highly stable (CV 2.5% - 5.5%).
- ▶ **Anomaly:** Column 3 exhibited a variance spike (8.54% CV), a hallmark of borderline transition to near-lethal antibiotic stresses.



10. Intra-Disk Region Detection

- ▶ **Method:** K-Means clustering ($K = 2$) spatially *within* pixels of sample wells to isolate 'Reacted' (pink) from 'Unreacted' (blue).
- ▶ **Inference:** Growth isn't a uniform color fade. Pink region area percentage expands non-linearly, showing localized metabolic activity.



Conclusion

- ▶ Successfully automated MIC identification using robust, multi-channel colorimetric analysis.
- ▶ Advanced ML (PCA, K-Means) revealed deeper color channel insights & non-linear behavioral responses.
- ▶ Spatial segmentation within discs proves growth is localized before diffusing.
- ▶ Early prediction models suggest feasibility of significantly reducing required incubation time.

Consolidated Results and Practical Takeaways

- ▶ **Main result:** Strong visual change but weak statistically significant a^* trend.
- ▶ **Most likely bottlenecks:** Image acquisition inconsistency, ROI alignment uncertainty, and illumination artifacts (large L^* drift).
- ▶ **Recommendation 1:** Standardize imaging geometry and camera settings across all timepoints.
- ▶ **Recommendation 2:** Validate ROI placement on every frame or switch to segmentation-first extraction.
- ▶ **Recommendation 3:** Keep percentile and ratio metrics as secondary diagnostics.
- ▶ **Recommendation 4:** Extend incubation time and/or strengthen indicator chemistry for larger biological effect size.

Road Ahead

- ▶ Reassess the validity and repeatability of the experiment with the mentors.
- ▶ Investigate the biology (eg. mode of action of the antibiotic) to model the bacterial–dye–antibiotic interaction as a reaction.
- ▶ Test hybrid methods.